Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	Team ID	Project Name	Maximum Marks
7 October 2026		Car Rental Request Automation in ServiceNow	4 Marks

Technical Architecture

The entire solution runs on the ServiceNow platform using the Service Catalog, Flow Designer, Business Rules and standard tables. No external custom servers are needed.

High-Level Architecture Overview

- User interface layer: Service Catalog form plus Service Portal / My Requests views
- Process and automation layer: Flow Designer (primary), Business Rules, Client Scripts
- Data layer: sc_request, sc_req_item, sc_task, sysapproval_approver and an optional custom Car Inventory table
- Notification layer: ServiceNow notification and email engine
- Deployment: Update Sets move configuration from Dev to Test to Prod
- Infrastructure: fully cloud-hosted ServiceNow instance (SaaS)

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User interface	Service Catalog item form for submitting requests Service Portal / My Requests for tracking status Back-end forms for approvers and the Transport team	ServiceNow Service Catalog Service Portal UI Policies / Client Scripts
2	Application logic – flow	End-to-end automation: trigger on RITM creation → manager approval → task creation → notifications → closure	ServiceNow Flow Designer Flow triggers and actions
3	Application logic – rules	Auto-fill fields, enforce mandatory data, update states after approval or task closure, set assignment groups	Business Rules Script Includes (if needed)
4	Client scripting	Form validation, dynamic field visibility, onChange / onSubmit logic	Client Scripts UI Policies Catalog Client Scripts
5	Database / tables	Request header and item (REQ / RITM) Fulfillment tasks (SCTASK) Approvals Optional Car Inventory custom table	sc_request, sc_req_item sc_task sysapproval_approver Custom table (u_car_inventory)
6	Notifications	Email and in-platform alerts on submission, approval, rejection, task assignment and closure	ServiceNow Notifications Email Templates Event Queue

S.No	Component	Description	Technology
7	SLA & reporting	SLA definitions for approval and fulfillment stages Dashboards and reports on cycle time and open requests	SLA Definitions Performance Analytics / Reports
8	Deployment	All configuration captured and moved with Update Sets	Update Sets Retrieved Update Sets
9	Infrastructure	Fully managed ServiceNow cloud instance No local servers needed	ServiceNow SaaS (cloud) Dev / Test / Prod instances

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Platform frameworks	Uses ServiceNow's out-of-the-box Service Catalog, Flow Designer and core tables, keeping custom code to a minimum	ServiceNow Platform Flow Designer JavaScript (server and client)
2	Security	Role-based ACLs on the catalog item, RITM, SCTASK and inventory table Managers approve only their direct reports' requests Audit history kept on all records	ACLs Roles and Groups sys_audit
3	Scalable architecture	Three logical tiers (UI / process / data) within ServiceNow Flow Designer scales with instance capacity New car types, locations or approval levels are easy to add	ServiceNow multi-instance architecture Flow Designer Update Sets
4	Availability	Inherits ServiceNow platform high availability and SLA No custom single points of failure	ServiceNow Cloud HA Instance redundancy
5	Performance	Lightweight catalog form and client scripts Efficient flow steps and GlideRecord usage Notifications queued asynchronously	Client Script best practices Flow Designer Event Queue

References

- ServiceNow Service Catalog and Flow Designer documentation – https://docs.servicenow.com/
- C4 model for architecture diagrams – https://c4model.com/
- ServiceNow Update Set best practices